

## SAFETY DATA SHEET

Creation Date 26-March-2009

Revision Date 01-September-2023

Revision Number 3

### 1. Identification

**Product Name** 2-Aminoethanethiol hydrochloride

**Cat No. :** SB00733DA; SB00733EA; SB00733ZZ

**CAS-No** 156-57-0  
**Synonyms** Cysteamine hydrochloride; 2-Mercaptoethylamine hydrochloride

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Fisher Scientific  
One Reagent Lane  
Fair Lawn, NJ 07410

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                   |            |
|-----------------------------------|------------|
| Acute oral toxicity               | Category 4 |
| Serious Eye Damage/Eye Irritation | Category 2 |
| Skin Sensitization                | Category 1 |

#### Label Elements

##### **Signal Word**

Warning

##### **Hazard Statements**

Harmful if swallowed  
May cause an allergic skin reaction  
Causes serious eye irritation

**Precautionary Statements****Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF SWALLOWED: Call a POISON CENTER/doctor if you feel unwell  
IF ON SKIN: Wash with plenty of soap and water  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Rinse mouth  
If skin irritation or rash occurs: Get medical advice/attention  
If eye irritation persists: Get medical advice/attention  
Take off contaminated clothing and wash it before reuse

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Stench

### 3. Composition/Information on Ingredients

| Component                  | CAS-No   | Weight % |
|----------------------------|----------|----------|
| Mercaptamine hydrochloride | 156-57-0 | >95      |

### 4. First-aid measures

|                                        |                                                                                                                                                                                                                                                      |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | If symptoms persist, call a physician.                                                                                                                                                                                                               |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                      |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                                                                    |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                                                                                         |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                                                                                                                                |
| <b>Most important symptoms/effects</b> | None reasonably foreseeable. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                |

### 5. Fire-fighting measures

|                                         |                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                              |
| <b>Flash Point</b>                      | No information available                                                              |
| <b>Method -</b>                         | No information available                                                              |
| <b>Autoignition Temperature</b>         | No information available                                                              |
| <b>Explosion Limits</b>                 |                                                                                       |
| <b>Upper</b>                            | No data available                                                                     |
| <b>Lower</b>                            | No data available                                                                     |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                              |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                              |

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>). Sulfur oxides. Hydrogen chloride gas.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
1

**Instability**  
1

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                              |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.            |
| <b>Environmental Precautions</b>            | Should not be released into the environment.                                                                 |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.                  |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Keep refrigerated. Store under an inert atmosphere. Protect from moisture. Incompatible Materials. Strong oxidizing agents. |

## 8. Exposure controls / personal protection

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b>  | This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.                                                                                                                                                                                                                                                                                                          |
| <b>Engineering Measures</b> | Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source |

**Personal protective equipment**

**Eye Protection**  
**Hand Protection**Goggles  
Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material                                      | Breakthrough time                 | Glove thickness | Glove comments         |
|-----------------------------------------------------|-----------------------------------|-----------------|------------------------|
| Nitrile rubber<br>Neoprene<br>Natural rubber<br>PVC | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

No protective equipment is needed under normal use conditions.

**Environmental exposure controls**

Do not allow material to contaminate ground water system.

**Hygiene Measures**

When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing.

## 9. Physical and chemical properties

|                                               |                             |
|-----------------------------------------------|-----------------------------|
| <b>Physical State</b>                         | Solid                       |
| <b>Appearance</b>                             | White                       |
| <b>Odor</b>                                   | Stench                      |
| <b>Odor Threshold</b>                         | No information available    |
| <b>pH</b>                                     | 3.3-5.0 400 g/L             |
| <b>Melting Point/Range</b>                    | 66 - 70 °C / 150.8 - 158 °F |
| <b>Boiling Point/Range</b>                    | No information available    |
| <b>Flash Point</b>                            | No information available    |
| <b>Evaporation Rate</b>                       | Not applicable              |
| <b>Flammability (solid,gas)</b>               | No information available    |
| <b>Flammability or explosive limits</b>       |                             |
| Upper                                         | No data available           |
| Lower                                         | No data available           |
| <b>Vapor Pressure</b>                         | No information available    |
| <b>Vapor Density</b>                          | Not applicable              |
| <b>Specific Gravity</b>                       | No information available    |
| <b>Solubility</b>                             | Soluble in water            |
| <b>Partition coefficient; n-octanol/water</b> | No data available           |
| <b>Autoignition Temperature</b>               | No information available    |
| <b>Decomposition Temperature</b>              | 150 °C                      |
| <b>Viscosity</b>                              | Not applicable              |
| <b>Molecular Formula</b>                      | C2 H7 N S . H Cl            |
| <b>Molecular Weight</b>                       | 113.61                      |

## 10. Stability and reactivity

**Reactive Hazard**

None known, based on information available

**Stability**

Hygroscopic. Sensitive to air.

**Conditions to Avoid**

Incompatible products. Excess heat. Avoid dust formation. Exposure to air. Exposure to moisture. Exposure to moist air or water.

|                                         |                                                                                                                                   |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Incompatible Materials</b>           | Strong oxidizing agents                                                                                                           |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Nitrogen oxides (NO <sub>x</sub> ), Sulfur oxides, Hydrogen chloride gas |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                          |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                                     |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

**LD50 Oral VALUE** 642 mg/kg (rat)

#### Component Information

| Component                  | LD50 Oral                             | LD50 Dermal | LC50 Inhalation |
|----------------------------|---------------------------------------|-------------|-----------------|
| Mercaptamine hydrochloride | 642 mg/kg (Rat)<br>1352 mg/kg (Mouse) | Not listed  | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** Irritating to eyes, respiratory system and skin

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                  | CAS-No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------------|----------|------------|------------|------------|------------|------------|
| Mercaptamine hydrochloride | 156-57-0 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** Mutagenic effects have occurred in experimental animals.

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** No information available.

**Mobility** . Will likely be mobile in the environment due to its water solubility.

| Component                  | log Pow |
|----------------------------|---------|
| Mercaptamine hydrochloride | -2.14   |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN3335  
 Proper Shipping Name AVIATION REGULATED SOLID, N.O.S.  
 Technical Name Mercaptamine hydrochloride  
 Hazard Class 9  
 Packing Group III

#### TDG

UN-No UN3335  
 Proper Shipping Name AVIATION REGULATED SOLID, N.O.S.  
 Hazard Class 9  
 Packing Group III

#### IATA

UN-No UN3335  
 Proper Shipping Name Aviation regulated solid, n.o.s.  
 Hazard Class 9  
 Packing Group III

**IMDG/IMO** Not regulated

### 15. Regulatory information

#### International Inventories

| Component                  | CAS-No   | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|----------------------------|----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Mercaptamine hydrochloride | 156-57-0 | X   | -    | X    | ACTIVE                                        | 205-858-1 | -      | -   |

| Component                  | CAS-No   | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|----------------------------|----------|-------|----------|------|------|------|------|-------|-------|
| Mercaptamine hydrochloride | 156-57-0 | X     | KE-01332 | -    | -    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

#### Other International Regulations

Authorisation/Restrictions according to EU REACH

Not applicable

Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                  | CAS-No   | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|----------------------------|----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Mercaptamine hydrochloride | 156-57-0 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                  | CAS-No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Mercaptamine hydrochloride | 156-57-0 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

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**Creation Date**

26-March-2009

**Revision Date**

01-September-2023

**Print Date**

01-September-2023

**Revision Summary**

This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**